

Investigation of Endurance Characteristics in 3-D NAND Flash Memory With Trap Profile Analysis

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Abstract—In this article, the endurance characteristic of 3-D NAND Flash memory is investigated. The bitline (BL) current and the word-line (WL) current are measured for erase/write (EW) cycling up to 50k. Then, the trap profiles in the bandgap-engineered tunneling oxide (BE-TOX) layer are extracted through trap spectroscopy by charge injection and sensing (TSCIS) technique and technology computer-aided design (TCAD) simulation. The trap profiles show that the traps in the BE-TOX layer are generated toward the charge trap layer (CTL) side to the poly-Si channel side as EW cycling increases. Also, it shows that the deep traps below 2.0 eV start to be generated after 10k EW cycling and a double Gaussian distribution is fit well. Through the trap profile results, the degradation of both the BL current and the WL current according to EW cycling is completely explained.

Index Terms—3-D NAND flash memory, bandgap-engineered tunneling oxide (BE-TOX) degradation, endurance, erase/write (EW) cycling, technology computer-aided design (TCAD) simulation, trap profiles, trap spectroscopy by charge injection and sensing (TSCIS).

I. INTRODUCTION

NAND flash memory is a popular type of non-volatile memory that is widely used in various electronic devices, such as smartphones, digital cameras, and solid-state drives (SSDs). Due to the scaling limits and reliability issues in 2-D planar NAND flash memory, a 3-D structure has become mainstream in NAND flash memory [1], [2], [3], [4].

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Three-dimensional NAND Flash memory consists of a poly-Si channel, a charge trap layer (CTL) as a storage layer, and a bandgap-engineered tunneling oxide (BE-TOX) layer to enhance erase performance [5]. Additionally, the use of multilevel cell (MLC) technology has become essential in order to reduce bit costs. One of the key challenges in the development of 3-D NAND flash memory is achieving high endurance and reliability, which is critical for the data errors and cell characteristics. Endurance refers to the ability of a memory cell to withstand repeated erase/write (EW) cycling without degradation. During EW cycling, Fowler–Nordheim (FN) tunneling current is dominant and can lead to degradation by the anode hole injection (AHI), resulting in the generation of bulk traps in the BE-TOX layer and interface traps in poly-Si channel/BE-TOX interface [6], [7], [8], [9]. This degradation can cause significant problems, such as a shift in the mid-gap voltage (V_{MG}), representing the voltage that is not affected by the interface traps in poly-Si channel/BE-TOX interface, subthreshold swing (S.S.) increase, and transconductance (g_m) decrease [10], [11].

Previous studies have extensively investigated the endurance characteristics of 2-D and 3-D NAND flash memory [9], [10], [11], [12], [13], [14], [15], [16], [17]. However, there is a lack of analysis of cell characteristics such as bitline (BL) and word-line (WL) current in memory cells subjected to high EW cycling above 10k. In this article, $I_{BL}-V_{WL}$ and $I_{WL}-V_{WL}$ curves are measured and analyzed during the EW cycling stress. The trap profiles in the BE-TOX layer are quantitatively extracted through the technology computer-aided design (TCAD) simulation and trap spectroscopy by charge injection and sensing (TSCIS) technique [18], [19]. Using the trap profiles, TCAD simulation, and energy band diagrams, the V_{MG} in the BL current and fully trapped voltage (V_{FT}), representing the voltage at which the dielectric layers above the poly-Si channel are fully trapped, as well as the tunneling current in the WL current is demonstrated intuitively. In addition, the trap generation as a function of the trap energy level is investigated during EW cycling.

II. EXPERIMENT SETUP

Fig. 1(a) shows the 3-D CTL NAND flash memory test element group (TEG) structure, where around 100 000 strings

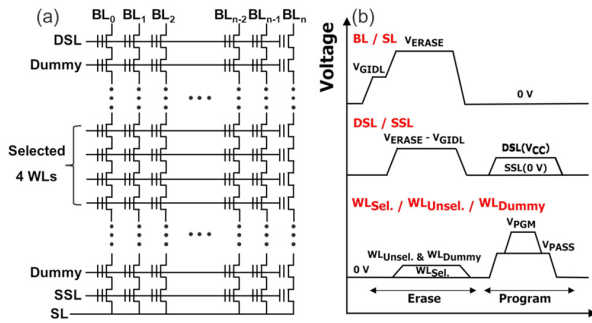


Fig. 1. (a) TEG of 3-D CTL NAND flash memory measured in the experiment. (b) Waveforms for EW cycling stress.

are connected in parallel. The string in TEG consists of BL, drain select line (DSL), WLs, dummy WLs, source select line (SSL), and source-line (SL). To measure the very small WL current, four WLs are connected to one pad as a selected WL. The endurance test is conducted in the middle of the string at room temperature (25°). EW cycling is run up to 50k and $I_{BL}-V_{WL}$ and $I_{WL}-V_{WL}$ curves are measured in fresh, 0.5k, 1k, 2k, 3k, 5k, 10k, 20k, 30k, 40k, and 50k EW cycling. The threshold voltage (V_{th}) is extracted by the constant current method. Fig. 1(b) shows the waveforms used in the EW cycling test. The gate-induced drain leakage (GIDL) erase operation and one pulse program operation are adjusted. During the erase operation, a small amount of positive voltage is applied to unselected and dummy WLs to prevent erase stress during the EW cycling.

III. I-V CURVE AND TSCIS MEASUREMENT

A. BL and WL Current Measurement Results

Fig. 2 shows the measurement results of BL current and WL current during 50k EW cycling. Fig. 2(a) and (b) show the $I_{BL}-V_{WL}$ curve of the erase state and program state. It shows the same results as our group's previous research that the V_{MG} is constant in the program state but shifts positively in the erase state due to the bulk trap generation in the BE-TOX layer during EW cycling [20]. Fig. 2(c) and (d) show the $I_{WL}-V_{WL}$ curve of the positive and negative sweep of the WL voltage during 50k EW cycling. The positive sweep of the WL current is measured at the erase state after each EW cycling, and the negative sweep of the WL current is measured after the positive sweep. The WL current shows the tunneling current following the WL bias. In Fig. 2(c), region A is when the low positive voltage is applied where the trap-assisted-tunneling (TAT) is dominant. Region B is when the middle positive voltage is applied where the modified FN tunneling [21] is dominant. The tunneling electrons are trapped in the BE-TOX layer, CTL, and blocking oxide (BOX) layer, leading to a saturation of electron tunneling current due to the saturation of the vertical field, even with an increase in WL voltage. Region C is when the high positive voltage is applied where the BE-TOX and CTL layers are fully trapped by the tunneling electrons. Since the dielectric layers above the poly-Si channel are fully trapped, the vertical field increases when the WL voltage increases, resulting in the electron tunneling current increases. The V_{FT} in region C is defined at a constant WL

current. In Fig. 2(d), region D is when the low negative voltage is applied where the TAT is dominant. Region E is when the middle negative voltage is applied where the modified FN tunneling is dominant and holes are tunneled and trapped in the BE-TOX and CTL layers. The hole tunneling current is saturated for the same reason as the electron tunneling current in region B. Region F is when the high negative voltage is applied where the electron back tunneling from the WL side is dominant.

In Fig. 2(a), V_{MG} shifts positively during EW cycling due to the bulk trap generation caused by the BE-TOX degradation. Similarly, in Fig. 2(c), V_{FT} shifts positively due to bulk trap generation caused by the BE-TOX degradation during EW cycling. However, in Fig. 2(c), the V_{FT} saturates after 10k EW cycling. In addition, unlike the V_{FT} , the TAT current in region A in Fig. 2(c) does not increase before 10k EW cycling and then increase after 10k EW cycling. Also, the negative sweep of WL voltage in Fig. 2(d) shows a different trend than in Fig. 2(c). These special trends of the WL currents should be analyzed not only with the trap density in the BE-TOX layer, but also with accurate trap profiles with the trap distance and trap level extracted through the TSCIS technique.

B. TCAD Simulation and TSCIS Measurement Results

Fig. 3(a) shows a schematic cross section of the 3-D NAND flash memory used in the TCAD simulation. The device parameters of the structure are the same as the TEG illustrated in Fig. 1(a). The BE-TOX layer consists of an oxide/nitride/oxide (O1/N1/O2) layer, CTL consists of a nitride layer, and BOX consists of oxide/high-K layers. In the poly-Si channel, grain boundaries (GBs) are assumed positioning under the center of each WL and spacer. As shown in Fig. 3(b), the U-shaped trap distribution is applied at each GB and uniform trap distribution is applied at poly-Si/BE-TOX and poly-Si/Filler oxide interfaces to calibrate measurement and TCAD simulation data [22]. Fig. 3(c) shows the fitting result of the virgin $I_{BL}-V_{WL}$ curve has been carried out successfully. Through the TCAD simulation, the energy band and electron density are obtained to calculate trap profiles from the TSCIS data.

The trap profiles in the BE-TOX layer during EW cycling are extracted through the TSCIS technique [18]. The trap energy level from the conduction band edge of each layer and trap distance from the poly-Si channel are extracted by varying the charge voltage (V_{CHRG}) and the charge time (t_{CHRG}). During the TSCIS charging time, the electrons are tunneled through direct tunneling and trapped in the BE-TOX layer. The trapped electrons change the energy band of the BE-TOX layer and tunneling probability. Therefore, to determine the trap position corresponding to the next t_{CHRG} , the trap position and trap density obtained from the previous t_{CHRG} are applied as unit interface trap density to reextract the energy band of the BE-TOX layer in the TCAD simulation.

The charging condition during the TSCIS should be selected to prevent trap generation in the BE-TOX layer. In this device, there are negligible differences in the discharge V_{th} and S.S. with initial V_{th} and S.S. even when V_{CHRG} is raised up to 7.0 V and t_{CHRG} is extended up to 2000 s for poly-Si

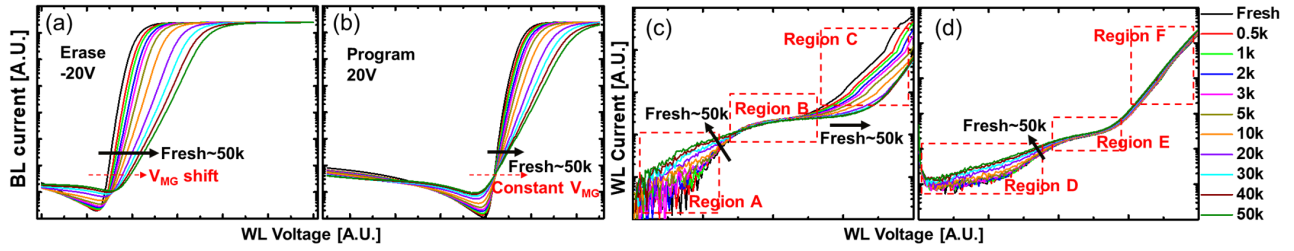


Fig. 2. Measurement results during 50k EW cycling. (a) I_{BL} - V_{WL} curve of the erase state. (b) I_{BL} - V_{WL} curve of the program state. (c) I_{WL} - V_{WL} curve during the positive sweep of the WL voltage. (d) I_{WL} - V_{WL} curve during the negative sweep of the WL voltage, respectively.

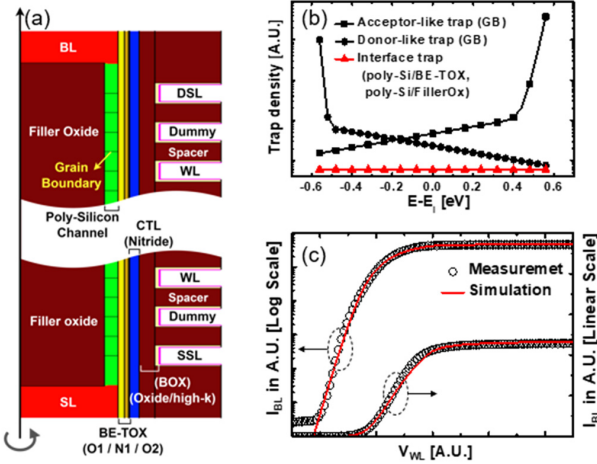


Fig. 3. (a) Schematic cross section of the 3-D CTL NAND flash memory used in the TCAD simulation. (b) U-shaped trap distribution of grain boundary in the poly-Si channel and interface traps of poly-Si/filler oxide and poly-Si/BE-TOX. (c) Fitting results of virgin BL current with experiment and TCAD simulation.

channel side TCSIS. There are negligible differences in the BL current before and after programming to 7.0 V. This indicates that there is negligible trap generation in both the BE-TOX layer and at the poly-Si/BE-TOX interface during the charging period. However, when the 7 V charging lasts 2000 s, soft programming-induced read disturbance occurs and causes a large V_{th} shift (ΔV_{th}). Therefore, the ΔV_{th} may not be measured when the V_{CHRG} is large or t_{CHRG} is long.

Fig. 4 shows the measurement results of the ΔV_{th} as a function of V_{CHRG} and t_{CHRG} for fresh, 1k, 3k, 5k, 10k, 30k, and 50k EW cycling, respectively. These results show the tendency of the bulk trap generation in the BE-TOX layer during EW cycling. In the fresh device, the initial I_{BL} - V_{WL} current is set near the virgin state to neglect the electron and hole trapping in the BE-TOX layer and CTL during the TSIS. Then the V_{CHRG} is applied to the WL during the 0.01 s–2000 s. At the end of each t_{CHRG} , the sense voltage ($V_{sense} < V_{th}$) is applied to measure the BL current. Through the measured BL current, ΔV_{th} is calculated based on the initial I_{BL} - V_{WL} curve. In the cycled device, the V_{MG} in the I_{BL} - V_{WL} curve shifts positively. Therefore, in order to set consistent initial conditions for both the fresh and cycled devices during TSCIS, it is necessary to align the V_{MG} of the cycled device with that of the fresh device. After that, the same procedure is conducted for the cycled device.

TABLE I
PARAMETERS USED FOR TSCIS CALCULATION

Symbol	Physical meaning
m_{O1}	Electron effective mass in O1 layer
m_{N1}	Electron effective mass in N1 layer
m_{O2}	Electron effective mass in O2 layer
σ_{O1}	Electron capture cross section in O1 layer
σ_{N1}	Electron capture cross section in N1 layer
σ_{O2}	Electron capture cross section in O2 layer

Fig. 5 shows the trap profiles for fresh, 1k, 3k, 5k, 10k, 30k, and 50k calculated from Fig. 4, respectively. The energy level of the traps is extracted from the conduction band edge of each layer, trap distance from the poly-Si channel is extracted through the filling factor (f_T) [19], and trap density is extracted from the ΔV_{th} at each trap position. The parameters used for the TSCIS calculation are listed in Table I. As shown in Fig. 5(b) and (c), the shallow traps with the trap energy levels of 0.9–1.5 eV start to be generated at the CTL side in the N1 layer during the low EW cycling. As shown in Fig. 5(d) and (e), the shallow traps and the deep traps below 2.0 eV are generated in the N1 layer from the CTL side to the poly-Si channel side when the EW cycling increase until the 10k. As shown in Fig. 5(f) and (g), the traps start to be generated in the O1/N1 interface when EW cycling increase after 10k.

IV. RESULTS AND DISCUSSION

A. Average Trap Density in BE-TOX Layer

Fig. 6 shows the average trap density of the BE-TOX layer ($N_{BE-TOX,AVG}$) extracted from the trap profiles in Fig. 5. Fig. 6(a) shows the trap distance from the poly-Si channel and $N_{BE-TOX,AVG}$ as a function of EW cycling extracted by moving average with 0.1 nm step. It intuitively shows that the traps in the BE-TOX layer are generated from the CTL side to the poly-Si channel side when EW cycling increases and the traps in the O1 layer start to be generated after the 10k EW cycling. Through these results, the TAT current in region A of Fig. 2(c) and region D of Fig. 2(d) can be demonstrated. Since the TAT current mechanism of region A and region D is the same, only the TAT current in region A is demonstrated in this section. Fig. 7 illustrates the TAT current flow in region A through the energy band diagram before and after 10k EW cycling. Since the positive sweep of the WL current is measured at the erase state after EW cycling, holes are trapped in the CTL

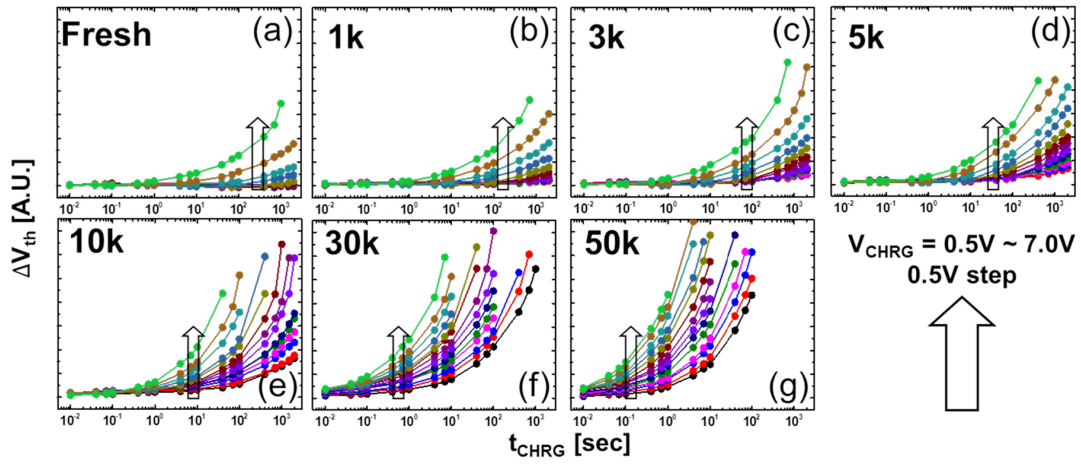


Fig. 4. Measurement results of ΔV_{th} as a function of V_{CHRG} and t_{CHRG} . (a) Fresh, (b) 1k, (c) 3k, (d) 5k, (e) 10k, (f) 30k, and (g) 50k EW cycling, respectively. The ΔV_{th} -axis scale is the same.

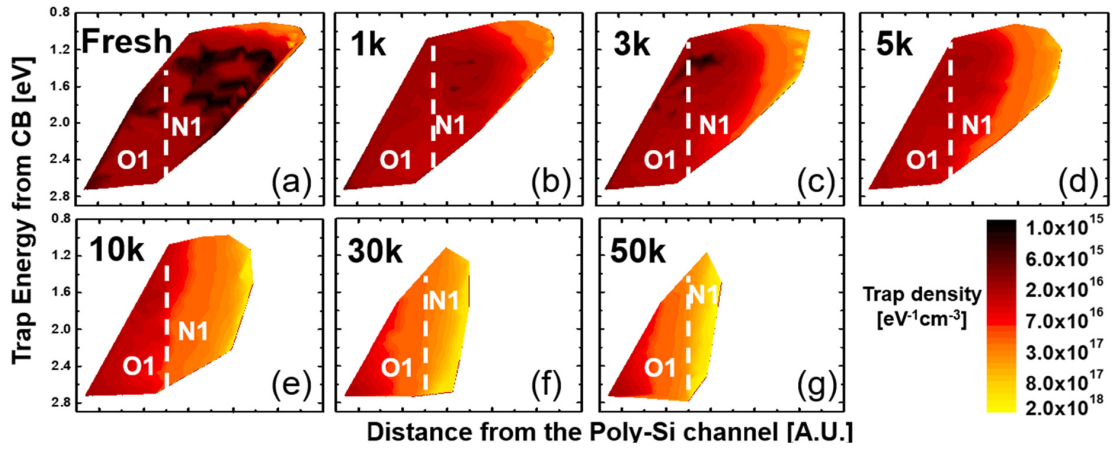


Fig. 5. Trap profiles of the BE-TOX layer. (a) Fresh, (b) 1k, (c) 3k, (d) 5k, (e) 10k, (f) 30k, and (g) 50k EW cycling, respectively.

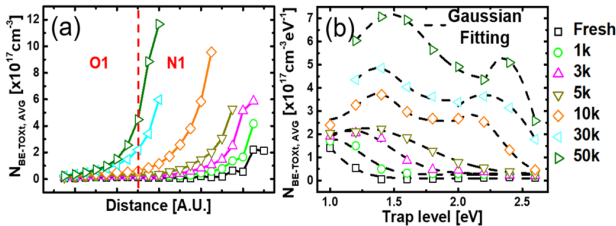


Fig. 6. (a) Average trap density as a function of trap distance from the poly-Si channel. (b) Average trap density as a function of the conduction band edge of each layer in the BE-TOX.

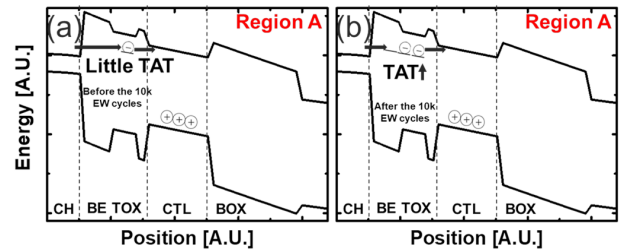


Fig. 7. Energy band diagram of 3-D NAND flash memory cell in region A (a) before and (b) after 10k EW cycling.

layer. For a more comprehensive understanding of the TAT current quantitatively, the number of traps and electrons has been presented as an arbitrary example. Due to the thickness of the dielectric layer stack in 3-D NAND Flash memory, electrons tunnel to the CTL layer rather than the WL, and then trapped in the CTL layer. Since a small WL voltage is applied in region A, the quantity of trapped electrons in the CTL layer due to TAT is very small and negligible. In Fig. 7(a), the TAT current is small and does not increase before 10k EW cycling because there is no trap generation near the O1/N1 interface which contribute most to the TAT current [24]. However, as shown in Fig. 7(b), the trap generation in the O1/N1 interface leads

to the increase in the TAT current in region A after 10k EW cycling.

Fig. 6(b) shows $N_{BE-TOX, AVG}$ as a function of the energy level from the conduction band edge of each layer extracted by moving average with 0.2 eV step. As illustrated in Fig. 6(b), a Gaussian distribution is fit well at each EW cycling before 10k EW cycling. Before 10k EW cycling, the peak trap density increases and energy level of the peak trap density shifts to the deep trap energy level. After 10k EW cycling, the peak trap density still increases and a double Gaussian distribution emerges at the deep trap energy level. This shows a significant

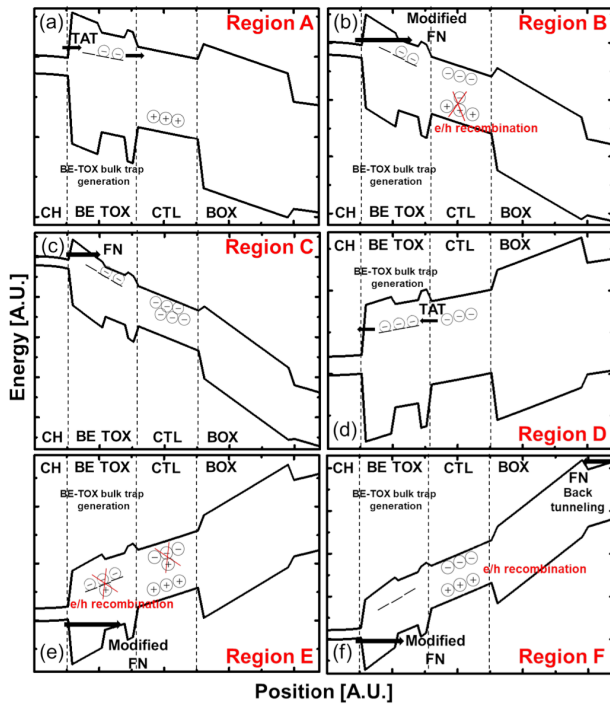


Fig. 8. Energy band diagram of 3-D NAND flash memory cell assuming after 10k EW cycling. (a) Region A. (b) Region B. (c) Region C. (d) Region D. (e) Region E. (f) Region F in Fig. 2, respectively.

increase in deep traps with energy levels below 2.0 eV after 10k EW cycling.

B. BL and WL Current Analysis With Energy Band Diagram

Fig. 8 depicts the energy band diagram of the 3-D NAND flash memory cell in each region of Fig. 2(c) and (d) assuming after 10k EW cycling. The traps in the BE-TOX layer start to be generated on the CTL side, and as EW cycling increases, they are generated toward the poly-Si channel side. Same as in Fig. 7, the number of traps and electrons/holes are presented as an arbitrary unit. Fig. 8(a) shows the energy band diagram of region A where the low positive voltage is applied. The TAT current in region A is demonstrated in Fig. 7. Fig. 8(b) shows the energy band diagram of region B where the middle positive voltage is applied. In the BE-TOX layer, the modified FN tunneling current flows and electrons recombine with holes and are trapped in the BE-TOX layer and CTL. Fig. 8(c) shows the energy band diagram of region C where the high positive voltage is applied. The traps generated in the CTL side BE-TOX layer can trap electrons. However, as shown in Fig. 8(b) and (c), the traps generated near the O1/N1 layer cannot trap electrons. This result is demonstrated by TCAD simulation and short-term TSCIS measurement. Fig. 9 shows the TCAD simulation results of electron trapping in the BE-TOX O1 layer depending on the trap energy level in region C. The nonlocal tunneling model [25] is considered and the trap density of the O1 layer is assumed as $1 \times 10^{18} \text{ cm}^{-3}$. The effective mass and capture cross section are listed in Table I, Huang–Rhys factor is 60, and phonon energy is 20 meV [24]. As shown in Fig. 9, since a higher WL voltage is applied

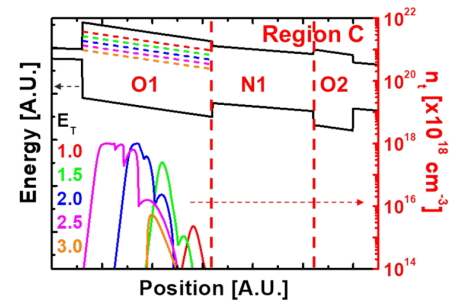


Fig. 9. TCAD simulation results of electron trapping in the BE-TOX O1 layer depending on the trap energy level in region C.

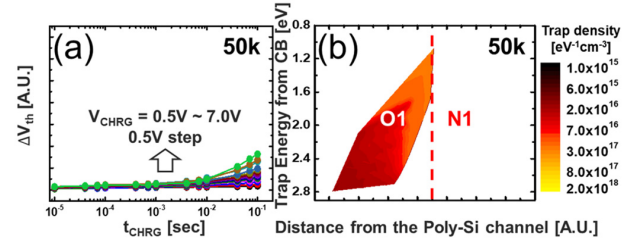


Fig. 10. (a) Measurement results of ΔV_{th} as a function of V_{CHRG} and t_{CHRG} in 50k EW cycling device. ΔV_{th} -axis scale is the same in Fig. 4. (b) Trap profiles of the BE-TOX layer in 50k EW cycling extracted from (a). The increment of the x-axis is the same in Fig. 5.

in region C compared to region A, TAT current increases, and electron trapping cannot be neglected. Consequently, electrons are trapped in the O1 layer due to TAT, even though the O1 layer is positioned along the FN tunneling path. The electron trapping due to TAT significantly depends on the trap level and position within the O1 layer. Fig. 10 shows the short term TSCIS measurement results with the range of t_{CHRG} is 10 μs to 0.1 s and the trap profile in BE-TOX layer in 50k EW cycling device. Since the t_{CHRG} is short, trap profile of BE-TOX layer closer to the poly-Si channel side can be obtained. As shown in Fig. 10(b), even though EW cycling increases to 50k, trap generation occurs at the O1/N1 interface, while trap generation in the BE-TOX layer near the poly-Si channel side remains negligible. Also, as shown in Fig. 9, tunneling electrons through TAT are predominantly trapped in the BE-TOX layer near the poly-Si channel side. Therefore, after 10k EW cycling, bulk traps are generated at the O1/N1 interface where electrons cannot be trapped. This leads to a constant density of fully trapped electrons and V_{FT} saturation.

As previously mentioned, the negative sweep is measured after the positive sweep. Therefore, the initial condition of the traps in the negative sweep is the same as in Fig. 8(c). Fig. 8(d) shows the energy band diagram of region D where the low negative voltage is applied. When the low negative voltage is applied, it is similar to when the GIDL pulse is applied in the GIDL erase operation, and the TAT flows. At this time, the electrons are trapped in the empty traps in the O1/N1 interface. Therefore, the total electrons trapped in the BE-TOX layer during the GIDL erase operation increase as EW cycling increases up to 50k. Fig. 8(e) shows the energy band diagram when the middle negative bias is applied, similar to when the erase pulse is applied in the GIDL erase operation. The holes

are tunneled through the modified FN tunneling current and recombine with the trapped electrons in the CTL and BE-TOX layer, and trapped in the CTL. Since the trapped electrons in the BE-TOX layer increase, the recombined holes in the BE-TOX layer increase, and the total trapped holes in the CTL layer decrease when EW cycling increases. Therefore, unlike the V_{FT} , the V_{MG} shifts positively both before and after 10k EW cycling. Fig. 8(f) shows the energy band diagram when the high negative voltage is applied. The holes are tunneled from the poly-Si channel through the modified FN tunneling current and electrons are tunneled from the WL through the FN tunneling (back tunneling) and electrons/holes are recombined in the CTL layer. Since the degradation in the BOX layer is negligible during EW cycling [17], the tunneling currents in region F are almost constant.

V. CONCLUSION

In this article, the impact of the BE-TOX layer degradation on the BL and WL current behavior of 3-D NAND flash memory cells during the endurance cycling was investigated. The $I_{BL}-V_{WL}$ and $I_{WL}-V_{WL}$ curves were measured, and the trap profiles were extracted through the TCAD simulation and TSCIS technique. The BL current showed that the V_{MG} shifted positively in the erase state due to the bulk trap generation in the BE-TOX layer during EW cycling. The WL current was dominated by the TAT, FN, and modified FN tunneling. In the WL current, the TAT current increased after 10k EW cycling. The V_{FT} shifted positively during EW cycling and then saturated after 10k EW cycling. These particular phenomena of the WL current were due to the fact that the trap generation in the BE-TOX layer occurs on the CTL side toward the poly-Si channel side during EW cycling demonstrated by the trap profiles and TCAD simulation. The different trends in the BL and WL currents were comprehensively explained through the energy band diagram. Also, trap profiles showed that the deep traps below 2.0 eV energy level were generated and a double Gaussian distribution emerged after 10k EW cycling. Additionally, since the TSCIS technique assumes SRH statistics and has limitations in extracting traps near the poly-Si channel/BE-TOX interface, ongoing research is required for a more accurate understanding of the trap profile.

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